

# Machine Learning Based Buyer Segmentation

## and Investment Profiling for Real Estate Market Intelligence

**Author:** Prince Bhardwaj    **Mentor:** Unified Mentor Program    **Date:**28 June 2026

Dataset: 2,000 Clients · 10,000 Transactions · Algorithm: K-Means Clustering (K = 4) · Tools: Python, scikit-learn, Streamlit

### 1. Abstract

This paper presents an AI-driven buyer segmentation system for **Parcl Co. Limited** using **K-Means clustering** on 7,305 merged client-transaction records. Four buyer segments were identified: *Global Investors*, *First-Time Buyers*, *Corporate Buyers*, and *Luxury Investors*. Each segment shows distinct patterns in price, property size, financing, and purchase intent, enabling Parcl to replace generic marketing with targeted, data-driven strategies.

**Keywords:** K-Means Clustering, Buyer Segmentation, Real Estate, Unsupervised ML, Investment Profiling

### 2. Introduction and Problem Statement

Real estate companies serve buyers with highly diverse motivations — from first-time homeowners to multinational corporate investors. Without segmentation, all buyers receive identical marketing, leading to wasted budget, poor conversion, and missed opportunities.

Parcl currently lacks a data-driven understanding of buyer types, investment motivations, geographic differences, and financing patterns. This project applies unsupervised machine learning to solve this by:

- Identifying hidden buyer clusters from transaction data
- Profiling each cluster with actionable business characteristics
- Delivering an interactive Streamlit dashboard for real-time monitoring

### 3. Dataset and Exploratory Data Analysis

#### 3.1 Data Sources

Two CSV files were merged on `client_id ↔ client_ref`, filtered to `listing_status = 'Sold'`, producing **7,305 records**.

Table 1: Dataset Summary

| File           | Rows   | Key Fields                                                                                                                              |
|----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------|
| clients.csv    | 2,000  | client_id, client_type, gender, country, region, date_of_birth, acquisition_purpose, satisfaction_score, loan_applied, referral_channel |
| properties.csv | 10,000 | listing_id, sale_price, floor_area_sqft, unit_category, listing_status, client_ref                                                      |

#### 3.2 Key EDA Findings

- Individual clients: **94.9%**
- Corporate clients: **5.1%**
- Investment purchases: **30.8%**
- Home purchases: **69.3%**
- Buyers with loan: **36.8%**
- Avg. sale price: **\$344,375**
- Price range: \$97K – \$737K
- Top country: **USA (76.9%)**
- Top region: **California (31.7%)**
- Countries represented: **20+**

Table 2: Referral Channel Distribution

| Channel         | Count | %     | Insight                         |
|-----------------|-------|-------|---------------------------------|
| Website         | 1,103 | 55.2% | Primary channel; invest in SEO  |
| Agency          | 705   | 35.3% | High-touch; suits luxury buyers |
| Client Referral | 192   | 9.6%  | Lowest cost; highest trust      |

## 4. Methodology

### 4.1 Step 1 — Data Cleaning

Cleaned `sale_price` (removed \$ and commas), parsed `date_of_birth` across three date formats, computed `age` as of June 2024, and imputed missing values with column medians. Result: 7,305 null-free rows.

### 4.2 Step 2 — Feature Encoding

Label Encoding was applied to five categorical variables:

Table 3: Categorical Encoding

| Feature                          | Encoded Values                      |
|----------------------------------|-------------------------------------|
| <code>client_type</code>         | Individual = 0, Company = 1         |
| <code>acquisition_purpose</code> | Home = 0, Investment = 1            |
| <code>loan_applied</code>        | No = 0, Yes = 1                     |
| <code>referral_channel</code>    | Agency = 0, Client = 1, Website = 2 |
| <code>unit_category</code>       | Apartment = 0, Office = 1           |

### 4.3 Step 3 — Feature Scaling

All 8 features were standardised with `StandardScaler` (zero mean, unit variance). This is critical — without scaling, `sale_price` (\$97K–\$737K range) would dominate `satisfaction_score` (1–5 range) in distance calculations.

**Final 8 features:** `age`, `satisfaction_score`, `sale_price_clean`, `floor_area_sqft`, `client_type_enc`, `acquisition_purpose_enc`, `loan_applied_enc`, `referral_channel_enc`

### 4.4 Step 4 — Optimal Cluster Selection

**K = 4 was selected** because it maps directly to four actionable business archetypes. While K = 6 yields a slightly higher silhouette score, the difference is negligible ( $\Delta = 0.003$ ) and interpretability matters more than marginal statistical gain.

### 4.5 Step 5 — K-Means Clustering

K-Means was run with `n_clusters=4`, `random_state=42`, `n_init=10` for stable, reproducible results.

Table 4: Elbow Method and Silhouette Scores

| K | Inertia | Silhouette | Note                         |
|---|---------|------------|------------------------------|
| 2 | 47,694  | 0.1798     | Too few groups               |
| 3 | 43,113  | 0.1648     | Insufficient detail          |
| 4 | 36,537  | 0.1883     | Selected — best business fit |
| 5 | 33,437  | 0.1804     | Marginal gain                |
| 6 | 31,008  | 0.1911     | Hard to interpret            |

## 5. Cluster Analysis and Segment Profiles

Table 5: All Four Segments at a Glance

| Segment                | Count | Avg Age | Avg Price        | Avg Area          | Loan% | Invest% |
|------------------------|-------|---------|------------------|-------------------|-------|---------|
| C1 – Global Investors  | 1,515 | 54.1    | \$275,395        | 917 sqft          | 37.0% | 100%    |
| C2 – First-Time Buyers | 2,892 | 54.7    | \$255,257        | 854 sqft          | 36.7% | 0%      |
| C3 – Corporate Buyers  | 381   | 45.9    | \$341,645        | 1,131 sqft        | 39.4% | 38.3%   |
| C4 – Luxury Investors  | 2,517 | 54.1    | <b>\$490,726</b> | <b>1,608 sqft</b> | 36.2% | 23.3%   |

**Key trait:** 100% investment-driven purchases. Individual buyers preferring compact units (917 sqft avg) for rental yield. Geographic spread: USA 77%, Canada 5.4%, UK 4.2%.

**Strategy:** ROI calculators, rental yield reports, international property expos, multi-currency payment options.

**Key trait:** 100% home purchases. Largest segment. Lowest avg price (\$255K) and smallest units (854 sqft). Website is primary discovery channel.

**Strategy:** Educational content, mortgage calculators, neighbourhood guides, virtual property tours, email drip campaigns.

**Key trait:** 100% company entities. Smallest count but highest per-buyer transaction volume. Mixed purpose: 38.3% investment, 61.7% operational. Highest loan rate (39.4%).

**Strategy:** Dedicated B2B account managers, bulk-purchase packages, LinkedIn outreach.

**Key trait:** Highest avg price (\$490K — 42% above dataset mean) and largest units (1,608 sqft). Mostly home purchase (76.7%). 63.8% purchase without a loan, indicating high net-worth buyers.  
**Strategy:** VIP showings, off-plan exclusives, concierge service, premium lifestyle marketing.

## 6. Dashboard and Recommendations

### 6.1 Streamlit Dashboard

An interactive dashboard (`app.py`) was built with four tabs: **Segment Overview**, **Investment Behavior**, **Geographic Analysis**, and **Segment Insights**. The sidebar provides five cascading filters (Country, Region, Acquisition Purpose, Client Type, Segment), where the **Region dropdown dynamically updates** based on the selected country. Deployed at: [share.streamlit.io](https://share.streamlit.io)

### 6.2 Strategic Recommendations

Table 6: Marketing Prioritisation by Segment

| Priority   | Segment                | Action                                         |
|------------|------------------------|------------------------------------------------|
| High       | C2 – First-Time Buyers | Invest in SEO and website UX; mortgage tools   |
| High       | C4 – Luxury Investors  | Premium brand, concierge, private showings     |
| Medium     | C1 – Global Investors  | International digital ads; ROI-focused content |
| High-Value | C3 – Corporate Buyers  | B2B account managers; bulk pricing packages    |

**Geographic Expansion:** California is saturated (31.7%). Nevada, Colorado, and Oregon show strong emerging buyer activity with lower competition.

**Pricing Tiers:** The \$235K spread between C2 (\$255K) and C4 (\$491K) justifies four distinct product tiers (Entry, Mid, Business, Premium) rather than uniform pricing.

#### Product Roadmap:

- **C1 & C4:** Investment Analytics Portal with rental yield and appreciation forecasts
- **C2:** First Home Buyer Journey tool with pre-qualification and neighbourhood maps
- **C3:** Corporate Portfolio Manager for multi-unit tracking and bulk invoicing

## 7. Conclusion

This project demonstrates that K-Means clustering on real estate transaction data reveals four statistically distinct and business-meaningful buyer segments. Each segment requires a different marketing message, product feature set, and pricing strategy.

By adopting segment-driven approaches, Parcl can reduce marketing waste, improve conversion rates, and capture higher-value clients — translating AI insights directly into measurable revenue growth.

Table 7: Technical Stack

| Component       | Technology                      | Purpose                 |
|-----------------|---------------------------------|-------------------------|
| Language        | Python 3.x                      | Core programming        |
| Data Processing | pandas, numpy                   | Cleaning and analysis   |
| ML Model        | scikit-learn KMeans             | Unsupervised clustering |
| Feature Prep    | StandardScaler,<br>LabelEncoder | Scaling and encoding    |
| Dashboard       | Streamlit                       | Interactive web app     |
| Deployment      | Streamlit Cloud                 | Public shareable URL    |